



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Wireshark Network Analysis Masterclass

Course Code: C1123

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 4.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
4.0	2 October 2026	Revised from the supplied v3 deck; current guidance, visuals and 18 self-contained labs.	Dr. Alfred Ang

TABLE OF CONTENTS

Course Information	4
Learning Outcomes	4
Learning Reinforcement	4
Course Schedule	4
Day 1 — Capture and analyst workflow.....	4
Day 2 — Time, statistics, filters and DNS.....	5
Day 3 — IP and transport diagnostics.....	5
Day 4 — Graphs, applications and reporting.....	6
Lab Reference (aligned to course topics)	7

Course Information

Course Title	Wireshark Network Analysis Masterclass
Course Reference	C1123
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	4 days · 7.5 instructional hours per day (30 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led demonstrations and practical labs; 30 instructional hours
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Plan scoped captures and configure a reproducible analyst profile.
- LO2: Navigate, filter, summarize and time network exchanges.
- LO3: Interpret DNS, ARP, IPv4, ICMP and UDP evidence.
- LO4: Diagnose TCP and application behaviour, including HTTP and authorised TLS inspection.
- LO5: Report observed facts, hypotheses and next actions with defensible evidence.

Learning Reinforcement

- Every topic is followed by hands-on labs that learners complete on their own machine.
- Each lab ends with a 'Test it' verification step so learners can confirm their own progress.
- The trainer circulates during lab time to give individual feedback and unblock learners.
- Each topic closes with a recap mapped back to the course learning outcomes.

Course Schedule

Day 1 — Capture and analyst workflow

Time	Duration	Topic / Activity
09:30–11:00	90 min	Concepts and demonstrations: topics 1, 2, 3, 4
11:00–11:15	15 min	Tea break
11:15–12:15	60 min	Lab 1: Establish a trace baseline; Lab 2: Plan capture placement; Lab 3: Create an analyst profile; Lab 4: Color and annotate evidence
12:15–13:00	45 min	Evidence interpretation and worked examples
13:00–14:00	60 min	Lunch

14:00–15:30	90 min	Continue labs: Lab 1: Establish a trace baseline; Lab 2: Plan capture placement; Lab 3: Create an analyst profile; Lab 4: Color and annotate evidence
15:30–15:45	15 min	Tea break
15:45–17:30	105 min	Worked examples and topic synthesis
17:30–18:00	30 min	Complete lab investigations and findings
18:00–18:30	30 min	Learning reinforcement, topic recap and reflection

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Day 2 — Time, statistics, filters and DNS

Time	Duration	Topic / Activity
09:30–11:00	90 min	Concepts and demonstrations: topics 5, 6, 7, 8, 9
11:00–11:15	15 min	Tea break
11:15–12:15	60 min	Lab 5: Separate path and server delay; Lab 6: Summarize traffic and a voice stream; Lab 7: Build a filter evidence matrix; Lab 8: Reconstruct an application dependency chain; Lab 9: Diagnose DNS failure and delay
12:15–13:00	45 min	Evidence interpretation and worked examples
13:00–14:00	60 min	Lunch
14:00–15:30	90 min	Continue labs: Lab 5: Separate path and server delay; Lab 6: Summarize traffic and a voice stream; Lab 7: Build a filter evidence matrix; Lab 8: Reconstruct an application dependency chain; Lab 9: Diagnose DNS failure and delay
15:30–15:45	15 min	Tea break
15:45–16:45	60 min	Worked examples and topic synthesis
16:45–18:00	75 min	Complete lab investigations and findings
18:00–18:30	30 min	Learning reinforcement, topic recap and reflection

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Day 3 — IP and transport diagnostics

Time	Duration	Topic / Activity
09:30–11:00	90 min	Concepts and demonstrations: topics 10,

		11, 12, 13, 14
11:00–11:15	15 min	Tea break
11:15–12:15	60 min	Lab 10: Verify link-local resolution; Lab 11: Classify IPv4 scope and headers; Lab 12: Interpret echo and unreachable messages; Lab 13: Follow datagrams and service refusal; Lab 14: Investigate retransmission and zero window
12:15–13:00	45 min	Evidence interpretation and worked examples
13:00–14:00	60 min	Lunch
14:00–15:30	90 min	Continue labs: Lab 10: Verify link-local resolution; Lab 11: Classify IPv4 scope and headers; Lab 12: Interpret echo and unreachable messages; Lab 13: Follow datagrams and service refusal; Lab 14: Investigate retransmission and zero window
15:30–15:45	15 min	Tea break
15:45–16:45	60 min	Worked examples and topic synthesis
16:45–18:00	75 min	Complete lab investigations and findings
18:00–18:30	30 min	Learning reinforcement, topic recap and reflection

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Day 4 — Graphs, applications and reporting

Time	Duration	Topic / Activity
09:30–11:00	90 min	Concepts and demonstrations: topics 15, 16, 17, 18
11:00–11:15	15 min	Tea break
11:15–12:15	60 min	Lab 15: Graph bytes, RTT and recovery; Lab 16: Inspect HTTP outcomes and HTTP/2 frames; Lab 17: Compare encrypted and decrypted views; Lab 18: Produce an evidence-led incident report
12:15–13:00	45 min	Evidence interpretation and worked examples
13:00–14:00	60 min	Lunch
14:00–15:30	90 min	Continue labs: Lab 15: Graph bytes, RTT and recovery; Lab 16: Inspect HTTP outcomes and HTTP/2 frames; Lab 17: Compare encrypted and decrypted views; Lab 18: Produce an evidence-led incident report

15:30–15:45	15 min	Tea break
15:45–16:30	45 min	Worked examples and topic synthesis
16:30–18:00	90 min	Complete lab investigations and findings
18:00–18:30	30 min	Learning reinforcement, topic recap and reflection

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Lab Reference (aligned to course topics)

Topic	Coverage	Labs
Topic 01: Introduction to Network Analysis and Wireshark	—	Lab 1
Topic 02: Capture Methods and Capture Filters	—	Lab 2
Topic 03: Global Preferences and Troubleshooting Profiles	—	Lab 3
Topic 04: Navigation and Coloring Techniques	—	Lab 4
Topic 05: Time Values and Delay Types	—	Lab 5
Topic 06: Trace Statistics and VoIP Overview	—	Lab 6
Topic 07: Display Filters	—	Lab 7
Topic 08: TCP/IP Communications and Resolution	—	Lab 8
Topic 09: DNS Traffic Analysis	—	Lab 9
Topic 10: ARP Traffic Analysis	—	Lab 10
Topic 11: IPv4 Traffic Analysis	—	Lab 11
Topic 12: ICMP Traffic Analysis	—	Lab 12
Topic 13: UDP Traffic Analysis	—	Lab 13
Topic 14: TCP Protocol Analysis	—	Lab 14
Topic 15: Traffic Graphs	—	Lab 15
Topic 16: HTTP and HTTP/2 Analysis	—	Lab 16
Topic 17: TLS-Encrypted Traffic Analysis	—	Lab 17
Topic 18: Ten Troubleshooting Steps and Reporting	—	Lab 18